

MedReserve

Project Name

MedReserve – Dynamic Medical Oxygen & Critical Supply System

Problem Statement

Critical healthcare facilities experience severe delays in acquiring life-saving supplies during emergency surges due to a lack of shared data transparency across regional hospital networks. Hospitals operating in silos are unable to signal their shortages to nearby facilities or central authorities, resulting in preventable deaths during public health crises.

Why does this problem exist?

Existing hospital management systems operate completely independently. There is no unified strategic or tactical platform for government bodies or suppliers to view collective regional shortages in real-time. Critical supply levels are still tracked manually or through disconnected spreadsheets in many facilities. There is no early warning system or inter-facility communication channel for supply emergencies, and procurement decisions are largely reactive rather than proactive.

Who are the Stakeholders?

Internal users:

1. Hospital Admin – manages facility-level inventory and procurement approvals
2. Chief Pharmacists – oversees medicine and oxygen stock levels
3. Emergency Ward Heads – first responders to critical shortage alerts
4. IT Department – maintains and monitors the system infrastructure

External users:

1. Bulk Medical Suppliers – receives alerts and dispatches supplies regionally
2. Regional Health Ministry Inspectors – monitors compliance and inventory status
3. Ambulance Networks – requires real-time supply routing for emergency dispatch
4. State Government Health Department – policy decisions and fund allocation

Expected system users:

1. Inventory Procurement Officers
2. Emergency Dispatchers
3. Supply Chain Coordinators

Objectives of the Project

1. Build a live operational tracker for critical medical supplies such as oxygen cylinders, IV fluids, blood units, and emergency medicines across a network of registered hospitals.
2. Automate reorder alerts for hospital administrators when local stock drops below a calculated tactical threshold — before a crisis occurs.
3. Create a strategic dashboard for health officials to view regional shortage maps and redistribute resources efficiently during a localized crisis.
4. Enable a supplier portal for bulk medical vendors to receive real-time shortage alerts and coordinate dispatch and delivery.
5. Maintain a complete audit trail of all procurement actions, approvals, and inter-facility supply transfers for compliance and government reporting.
6. Generate predictive shortage reports using historical consumption data to help hospitals prepare for seasonal outbreaks and public health emergencies.
7. Facilitate inter-hospital resource sharing by allowing surplus inventory at one facility to be transferred to another in need within the same region.
8. Provide role-based access control so that each user — hospital admin, pharmacist, supplier, or ministry official — sees only the data relevant to their function.

Expected Outputs

1. Zero unplanned stockouts of critical medical supplies during emergency surge periods.
2. Reduction in emergency procurement lead time by at least 60%.
3. Full real-time visibility of regional supply status for all health authorities.
4. Improved coordination between hospitals, suppliers, and the health ministry.
5. Faster inter-facility resource redistribution during localized crises.
6. Data-driven seasonal planning using predictive shortage analytics.

Hands-on Activity

Identify the following types of information from your project:

Operational Information

1. Live real-time tracking of oxygen cylinders, IV fluids, blood units, and emergency medicines.
2. Hospital Admin managing facility-level inventory and procurement approvals daily.
3. Chief Pharmacist overseeing medicine and oxygen stock levels.
4. Emergency Ward Heads responding to critical shortage alerts immediately.
5. Inventory Procurement Officers raising and processing reorder requests.
6. Emergency Dispatchers using real-time supply routing for ambulance networks.
7. Supplier response time to shortage alerts — target under 2 hours.
8. System uptime during emergencies — target 99.9%.

Tactical Information

1. Automated reorder alerts when stock drops below a calculated threshold — before a crisis occurs.
2. Inter-hospital resource sharing — surplus inventory at one hospital transferred to another in need.
3. Regional Health Ministry Inspectors monitoring compliance and inventory status.
4. Reduction in emergency procurement lead time by at least 60%.
5. Supplier portal for bulk vendors to receive shortage alerts and coordinate dispatch.
6. Role-based access control so each user sees only data relevant to their function.
7. Audit trail of all procurement actions and inter-facility transfers for compliance reporting.

Strategic Information

1. MedReserve aims to eliminate preventable deaths caused by medical supply shortages during public health crises.
2. Long-term integration into a national health emergency response framework for all of India.
3. Policy decisions and fund allocation by the State Government Health Department.
4. Strategic dashboard for health officials to view regional shortage maps across the entire network.
5. Zero unplanned stockouts of critical medical supplies during emergency surge periods.
6. Data-driven seasonal planning using predictive shortage analytics for future outbreaks.

Use Case Diagram

